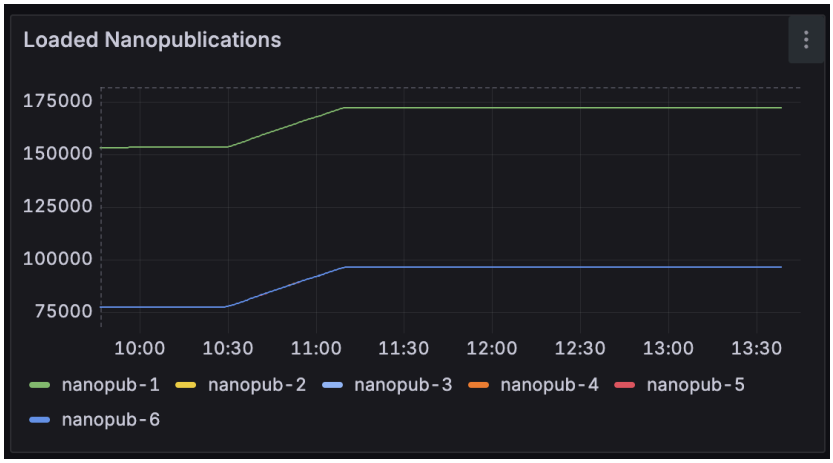
Initial run on test set (test mode on, w/o replication among registries), showed that 9 generators can publish **110k+** nanopublications in 40 minutes.



With each generator producing around 12k nanopublications, making the speed of a generator in this scenario be around **5 np/s**.

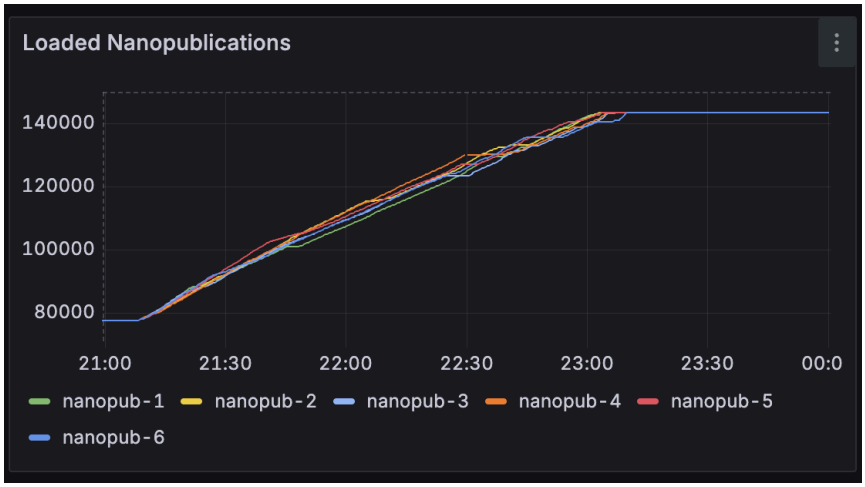
Main setup: 6 registries initialized with the content from the public network, Registry test mode was turned off.

Test design:

- 9 generators on 3 client devices (3 per device). Device: Raspberry Pi 5.
- Target: ~100k nanopubs submission
- Querying with 24 clients for 20 minutes for latency check

The test mode was turned off, allowing for registries to replicate received nanopublications.

9 generators were run for 36 minutes, resulting in **~65k+** submitted nanopublications:



The speed of nanopublication ingestion compared to the previous scenario is **~3.38 np/s.** Therefore, each generator produced ~7k nanopublications. Each generator is single-threaded, so if the Registry takes longer to ingest the nanopub, the generator will also slow down.

This illustrates that connected registries lead to a **1.47x** slower ingestion rate compared to standalone registries.

During the run of the generators, errors were observed:

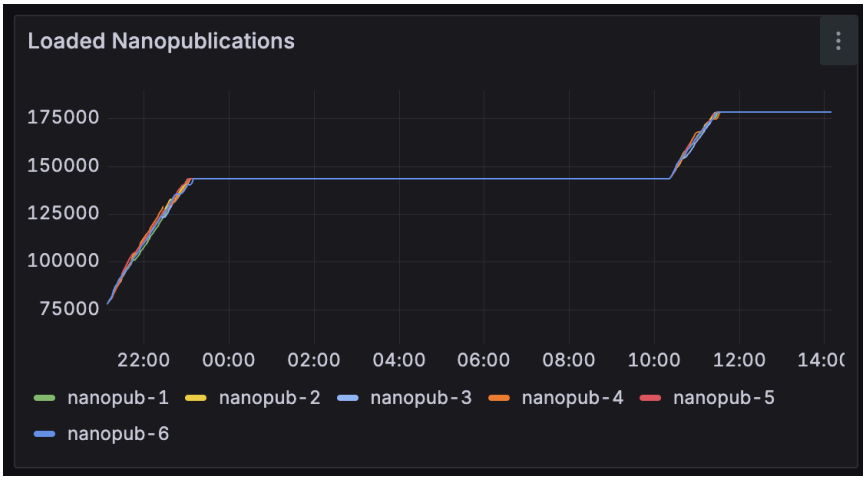
```
Published nanopub: http://purl.org/np/RARBgLGaxEW5sGUwCx22tG2rap95g8DLGHL7wqZz75Z90
Error publishing nanopub: 400 Client Error: Error processing nanopub: Write operation error on
server nanopub-2-registry-db:27017. Write error: WriteError{code=11000, message='E11000
duplicate key error collection: nanopubRegistry.nanopubs index: counter_-1 dup key: { counter:
79970 }', details={}}. for url: http://192.168.8.22:30782/
Published nanopub: http://purl.org/np/RAMM4vh5mQUvXweJqIvcequqDmg7MV6IdasTvXsbT7osQ
Published nanopub: http://purl.org/np/RA3ZwxeDmmZBbOcsHQVVITc8Sn-G0IBjQhOO0L1E2VrHU
Published nanopub: http://purl.org/np/RAikY6FIrDHkGwZbvlWf1yWmzlL3oV4ijloEqQRIAuq1A
Published nanopub: http://purl.org/np/RASx6PyvjqhRW1osP0ISSrljV1PzB5HTxOmcyCY6Do8n4
Published nanopub: http://purl.org/np/RAW55VnHPoeYHlMzdVCbGkStKUJKov_mPox5DDHIC7RfY
Published nanopub: http://purl.org/np/RA11Xwj8IhvRA7pk7NFIaArkX2F3lb4QkOL2e-wNs42go
Error publishing nanopub: 400 Client Error: Error processing nanopub: Write operation error on
server nanopub-1-registry-db:27017. Write error: WriteError{code=11000, message='E11000
duplicate key error collection: nanopubRegistry.nanopubs index: counter_-1 dup key: { counter:
80458 }', details={}}. for url: http://192.168.8.22:30782/
```

Text version of the error: Error publishing nanopub: 400 Client Error: Error processing nanopub: Write operation error on server nanopub-2-registry-db:27017. Write error: WriteError{code=11000, message='E11000 duplicate key error collection: nanopubRegistry.nanopubs index: counter_-1 dup key: { counter: 79970 }', details={}}. for url: http://192.168.8.22:30782/

The errors appeared on average 128 times per generator during the 36-minute run. **Error incidence rate: 1.75%**

We have not investigated the root cause of the problem, but it's likely a race condition in the Registry code. It's likely that the error will only show up under sustained, heavy load. We are not sure if this should be investigated before further experiments or not.

The total number of nanopublications was increased to ~178k:

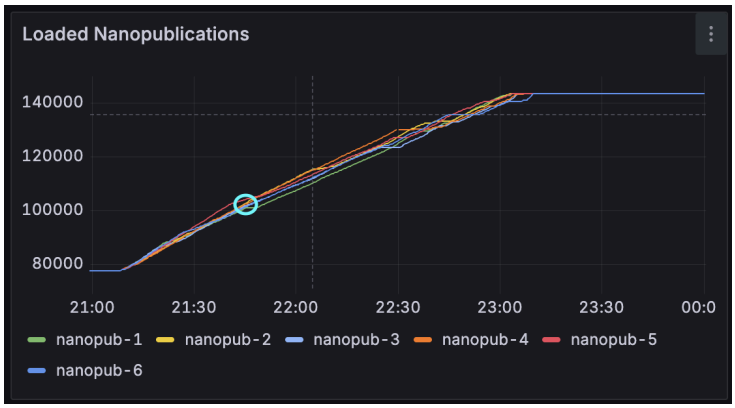


Querying with 24 clients (8 clients per device) was performed for 20 minutes after the registries finished replication.

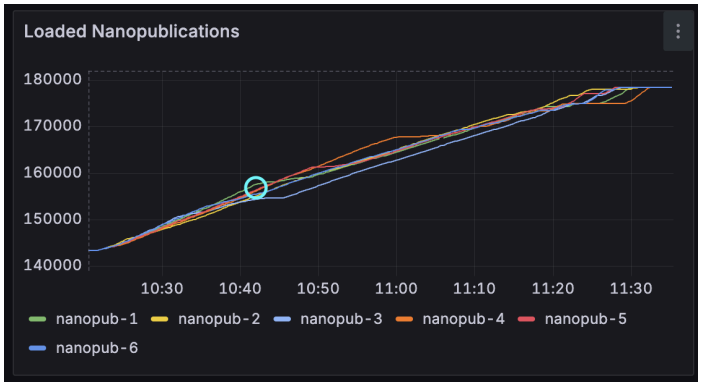
On average, 88 timeout messages were received per querying instance (3 instances, 50 clients each), which is a similar result to the one we obtained in December 2025 with a smaller network (~70k nanopubs).

Replication speed:

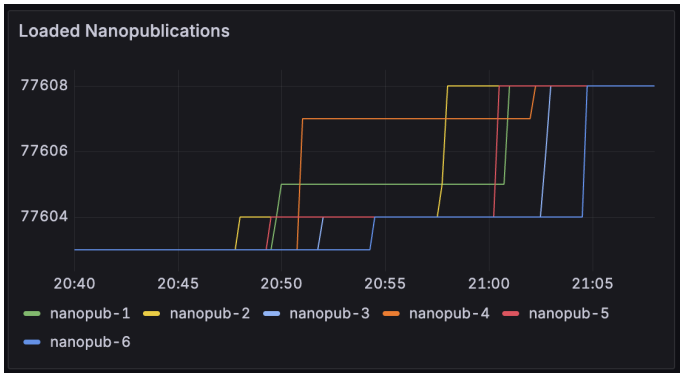
In the image below, the blue circle marks the approximate place where generators stopped submitting nanopublications (65k newly published nanopublications over 36 minutes). It is seen that registries keep syncing for **~1h 25m**.



The second batch of published nanopublications took 20 minutes for **35k** nanopublications. The overall time for synchronization **~50m.**

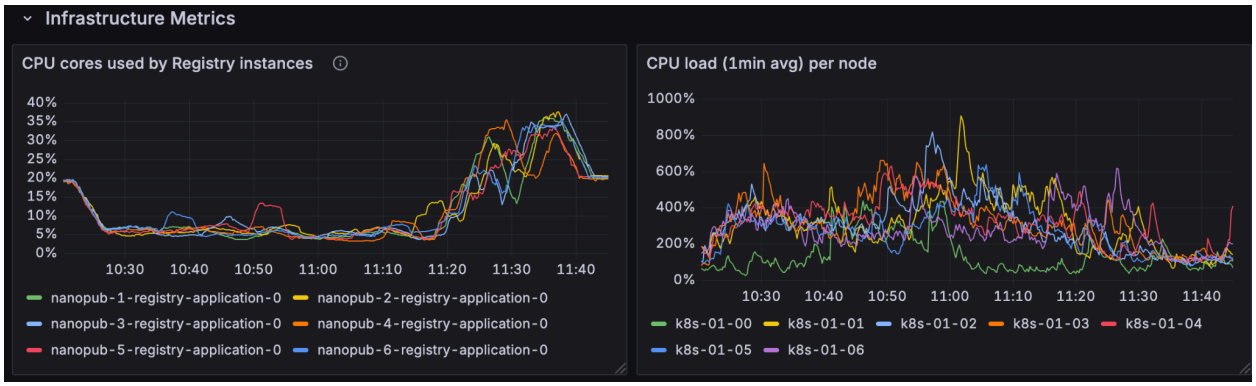


Previously, it was also observed that 6 nanopublications submitted over the course of 2 seconds took around 17 minutes to sync.



This replication rate is much slower than we expected, and is likely due to how the algorithm for nanopub replication works, and not due to hardware constraints.

CPU usage during the same period:



As can be seen, CPU usage during the slow replication was relatively small.